

Mini Project

Customer Sales Insights Pipeline using Azure Data Factory

Objective

The objective of this mini project is to build an end-to-end data pipeline using Microsoft Azure services. The project demonstrates how Azure Blob Storage and Azure Data Factory can be integrated to read a CSV file from cloud storage, validate its metadata, and copy it to another storage location successfully.

Problem Statement

Build a complete data pipeline that reads a CSV file stored in Azure Blob Storage, validates the file using the **Get Metadata** activity, and copies it to a new destination using **Azure Data Factory**.

Services Used

- Microsoft Azure Portal
- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)

Implementation

Step 1: Storage Configuration

A Storage Account was created in Azure, followed by the creation of two Blob Containers.

- **Source Container:** Stores the original Superstore CSV dataset.
- **Destination Container:** Stores the copied output file generated by the pipeline.

The Superstore dataset was uploaded successfully to the source container.

Step 2: Azure Data Factory Setup

An Azure Data Factory instance was created to build and manage the data pipeline.

The following components were configured:

- Linked Service to establish a connection with Azure Blob Storage.
- Source Dataset pointing to the uploaded CSV file.
- Destination Dataset pointing to the output location inside the destination container.

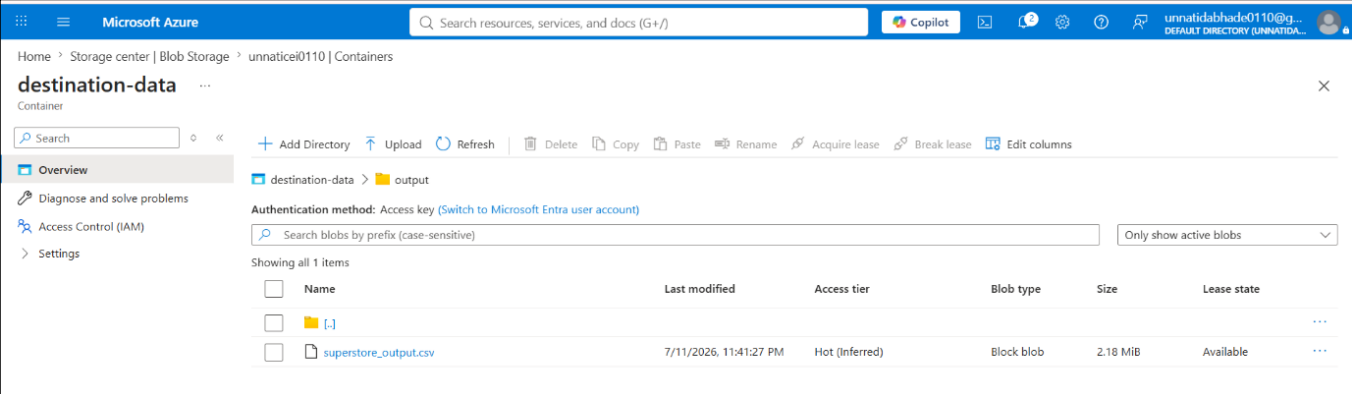
Step 3: Metadata Validation

The **Get Metadata** activity was added to the pipeline to validate the source file before processing.

This activity verifies important information about the file, such as:

- File existence
- File size
- Last modified date and time

Performing metadata validation ensures that the required source file is available before executing the data copy operation.



The screenshot displays the Microsoft Azure Storage Explorer interface. The top navigation bar shows the user is logged in as 'unnatidabhade0110@g...'. The main pane shows the 'destination-data' container, which contains a sub-container named 'output'. The 'output' container lists one item: 'superstore_output.csv'. The table below shows the details of this item.

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					...
superstore_output.csv	7/11/2026, 11:41:27 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Step 4: Data Pipeline

A pipeline named **CustomerSalesInsightsPipeline** was created.

The pipeline consists of two activities executed sequentially:

1. **Get File Metadata** – Validates the source CSV file.
2. **Copy Sales Data** – Copies the validated CSV file from the source container to the destination container.

*Since only a single CSV file was processed, a **ForEach** activity was not included, as it is mainly used for processing multiple files.*

The top part of the image shows a visual representation of the pipeline. It consists of two activities connected sequentially: 'Get Metadata' (containing 'Get File Metadata') and 'Copy data' (containing 'Copy Sales Data'). Both activity boxes have a green checkmark in the top right corner, indicating successful execution.

The bottom part of the image shows the 'Output' tab of the pipeline's execution details. It displays the following information:

- Pipeline ID:** deaffc3c-d58a-4b9a-bd
- run ID:** 1c-54c559d09a95
- Pipeline status:** Succeeded (indicated by a green checkmark)
- Buttons:** View debug run consumption, Monitor in Azure Metrics, Export to CSV
- Table:** A table showing the execution status of the activities.

Activity name	Activity st...	Activit...	Rt
Copy Sales Data	✓ Succeeded	Copy data	7/
Get File Metadata	✓ Succeeded	Get Metadata	7/

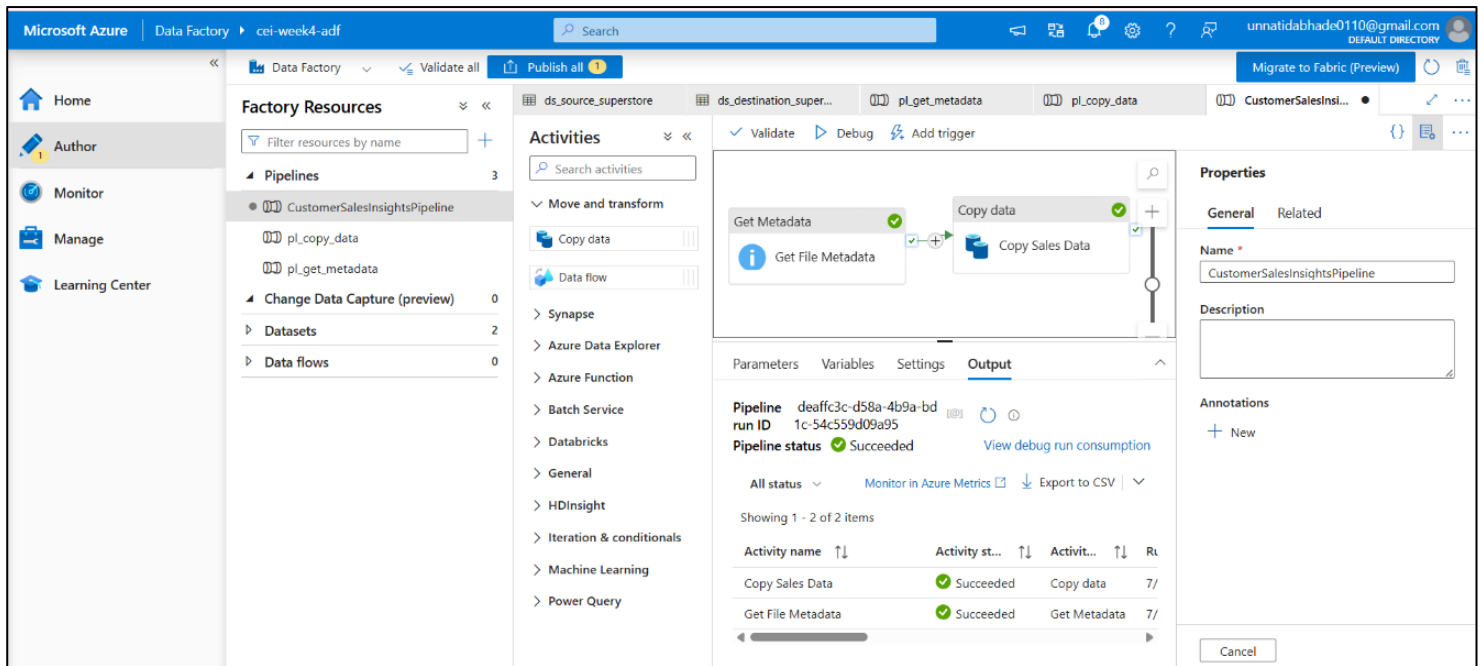
Step 5: Pipeline Execution

The pipeline was executed using the **Debug** option in Azure Data Factory.

Both activities completed successfully without any errors.

The execution status confirmed that:

- Metadata validation completed successfully.
- The CSV file was copied successfully.
- The pipeline execution status was **Succeeded**.



Results

The mini project was completed successfully.

The implemented pipeline achieved the following:

- Successfully connected Azure Blob Storage with Azure Data Factory.
- Retrieved metadata of the source CSV file.
- Validated the availability of the dataset before processing.
- Copied the source file to the destination container.
- Executed the complete pipeline successfully without any failures.

The output file **superstore_output.csv** was successfully created in the destination container.

Conclusion

This mini project provided practical experience in building an end-to-end cloud data pipeline using Microsoft Azure. It demonstrated the process of connecting cloud storage with Azure Data Factory, validating source data using the Get Metadata activity, and automating data movement through the Copy Data activity.

The project also helped in understanding the basic workflow of Azure Data Factory, including linked services, datasets, pipeline creation, execution, and monitoring. Overall, it strengthened the understanding of cloud-based data integration and ETL processes using Azure services.